

Proposal: Simulation of Morphy Drone Impacts Against a Rigid Surface

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Abstract

We propose to simulate and analyze collisions of the compliant multicopter *Morphy*[\[2\]](#) against rigid surfaces. Our model treats the vehicle as a multi body system composed of a central rigid body and four arm-motor-propeller assemblies connected through elastic joints. Each elastic joint is represented as a rotational spring-damper coupling that permits azimuth, elevation and torsional deflections while transmitting thrust forces and reaction torques. We will implement contact handling with multiple alternatives (impulse-based and IPC).



1 Objective

The goal of this project is to develop a physically based simulation framework that models the impact of a drone with a rigid surface. Specifically, we aim to reproduce the drone's fall, the collision with the ground, and the subsequent dynamic response of its rigid arms (bouncing, rotation, recoil).

- **Bonus:** Extend the scenario so that the drone does not simply go under free fall, but instead collides with another object. The collision should be able to transfer momentum to that object and potentially alter its motion.

2 System Overview

We model Morphy as:

- One central body frame $\{B\}$; four propeller frames $\{P_i\}$; and four joint frames $\{H_i\}$ at the elastic couplings.

- Each arm assembly (rigid arm + motor + propeller) is a rigid body with mass m_{P_i} and inertia J_{P_i} .
- Each elastic joint allows azimuth ϕ_i and elevation ϑ_i deflections and torsion χ_i and is modeled by a rotational spring-damper around local axes with stiffness $k_\phi, k_\vartheta, k_\chi$ and damping $c_\phi, c_\vartheta, c_\chi$.

3 Collision Handling

We will implement and compare:

- A.1 Impulse-based.** Instant velocity correction after predicted unconstrained motion.
- A.2 Variational penalty / IPC.** Formulate contact via smooth barrier potentials and solve via implicit integration with intersection-aware line search.

Assumptions & Simplifications

- No aerodynamic interaction, propeller wake or ground-effect modelling is considered.
- No closed-loop controller is implemented; we apply a fixed or scheduled thrust input/constant velocity on the main body and focus purely on collision dynamics.

4 Deliverables

1. Development of the multibody model of *Morphy*: definition of rigid bodies, compliant joints (rotational spring-dampers).
2. Integration of URDF (or equivalent) for visualization and verification of kinematics/geometry in a suitable viewer.
3. Implementation of a simple contact module: static plane contact only, basic impulse-based response, no friction.
4. Extension to A.2, comparison of the two methods.
5. **Bouns:** Collision with non-static objects.

Extra (if time permits and you guys like it)

If time permits, we will explore a unified articulated rigid-body simulation framework following the ideas of Weinstein, Teran, and Fedkiw [1], where contact, collision, and joint constraints are handled consistently in maximal coordinates through an impulse-based formulation with pre-stabilization of constraints. This would allow a fully integrated simulation of Morphy's compliant joints and collisions within a single solver, extending beyond separate contact and joint modules.

References

- [1] Weinstein, R., Teran, J., and Fedkiw, R. (2006). *Dynamic Simulation of Articulated Rigid Bodies with Contact and Collision*. IEEE Transactions on Visualization and Computer Graphics, 12(3):365–374. math.ucdavis.edu/~jteran/papers/WTF06.pdf

- [2] De Petris, P., Nissov, M., and Alexis, K. (2025). *Morphy: A Compliant and Morphologically Aware Flying Robot*. *Advanced Intelligent Systems*, 7(1):2400493. doi.org/10.1002/aisy.202400493